

Miles Reid NOTE

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1. MOTIVATION

This one is reference to Miles Reid Notes, for me to easily read and memorized in my mind. This one also summarization to Week 1 until Week 5 of my lecturer's consultation.

2. AFFINE VARIETIES

2.1 NOETHERIAN RINGS

Definition 2.1 (Noetherian). A ring R is called Noetherian if every ideal of R is finitely generated.

Theorem 2.2 (Hilbert Basis Theorem). *Let R be a Noetherian ring. Then the polynomial ring $R[x]$ is also Noetherian.*

3. ZARISKI TOPOLOGY

3.1 ALGEBRAIC SET

3.2 THE CORRESPONDING OF ALGEBRAIC SET

3.3 IRREDUCIBLE ALGEBRAIC SET

4. NULLSTELLENSATZ

4.1 RADICAL IDEALS

4.2 NULLSTELLENSATZ THEOREMS

Or also know as Hilbert's zeros theorem.

Theorem 4.1 (Weak Nullstellensatz, geometric form). *Suppose k is an algebraically closed field. Let I be a proper ideal in the polynomial ring $k[x_1, \dots, x_n]$. Then we have*

$$V(I) \neq \emptyset$$

i.e., there exists a point $(a_1, \dots, a_n) \in k^n$ such that $f(a_1, \dots, a_n) = 0$ for all $f \in I$.

Theorem 4.2 (Weak Nullstellensatz, algebraic form). *Suppose k is an algebraically closed field, and let $\mathfrak{m}$ is maximal ideal of $k[x_1, \dots, x_n]$, then we have*

$$\mathfrak{m} = (x_1 - a_1, \dots, x_n - a_n)$$

Theorem 4.3 (Strong Nullstellensatz). *Suppose k is an algebraically closed field. Let I be an ideal in the polynomial ring $k[x_1, \dots, x_n]$. Then the ideal of all polynomials that vanish on the variety defined by I is equal to the radical of I , i.e.,*

$$I(V(I)) = \sqrt{I}.$$

4.3 COROLLARY OF NULLSTELLENSATZ